6) a) If the sum of Right hand poles (+), clockwise encirclements of (-1,0) (+), and counter clockwise encirclements of (-1,0) (-) is >= 0 then the system is unstable. There are 2 RHP poles, and 1 N_{CCW}, ~~[illegible]~~ adding up to 1. So the closed loop system will be unstable.

---

b) $G_{OL}(s) = \frac{s^2 - 18s - 10}{s^2 - 5s + 4}$

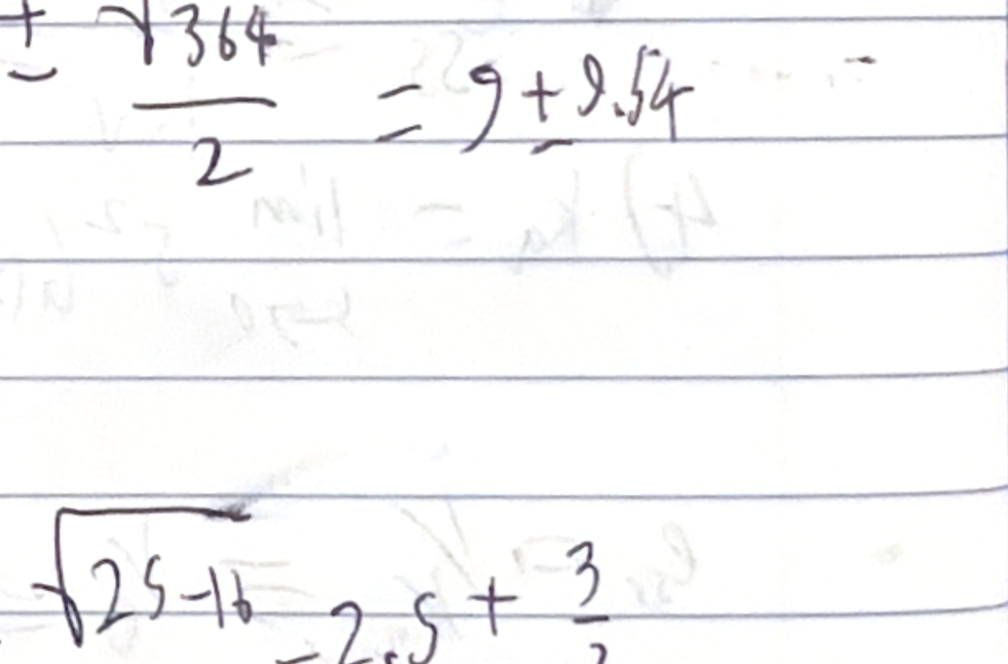
1) Zeros: $\frac{+18 \pm \sqrt{18^2 - 4(1)(-10)}}{2(1)} = 9 \pm \frac{\sqrt{364}}{2} = 9 \pm 9.54$

Zeros = 18.54, -0.54

2) $\frac{5 \pm \sqrt{(-5)^2 - 4(1)(4)}}{2(1)} = 2.5 \pm \frac{\sqrt{25-16}}{2} = 2.5 \pm \frac{3}{2}$

poles: 4, 1

3)



4) Both poles are positive ∴ The open loop system is unstable

c) $G_{CL}(s) = \dfrac{\dfrac{s^2-18s-10}{s^2-5s+4}}{1+\dfrac{s^2-18s-10}{s^2-5s+4}(1)} \cdot \dfrac{s^2-5s+4}{s^2-5s+4} = \dfrac{s^2-18s-10}{(s^2-5s+4)+(s^2-18s-10)}$

$= \dfrac{s^2-18s-10}{2s^2-23s-6}$

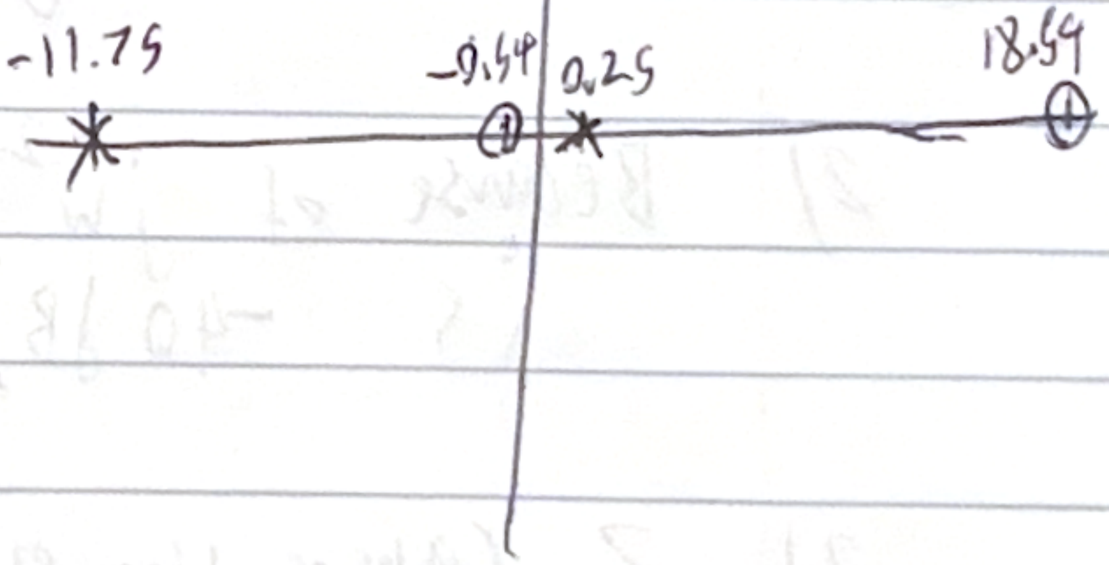
1) zeros are the same: **zero = 18.54, −0.54**

2) poles: $\dfrac{-23 \pm \sqrt{(-23)^2-4(2)(-6)}}{2(2)} = \dfrac{-23}{4} \pm \dfrac{\sqrt{577}}{4} = -5.75 \pm 6$

poles = 0.25, −11.75

3)

4 B) There is still a right hand pole ∴ the system is still unstable



6.

d)

```
>> NUM = [1 -18 -10]

NUM =

     1   -18   -10

>> DEN = [1 -5 4]

DEN =

     1    -5     4

>> sys = tf(NUM,DEN)

sys =

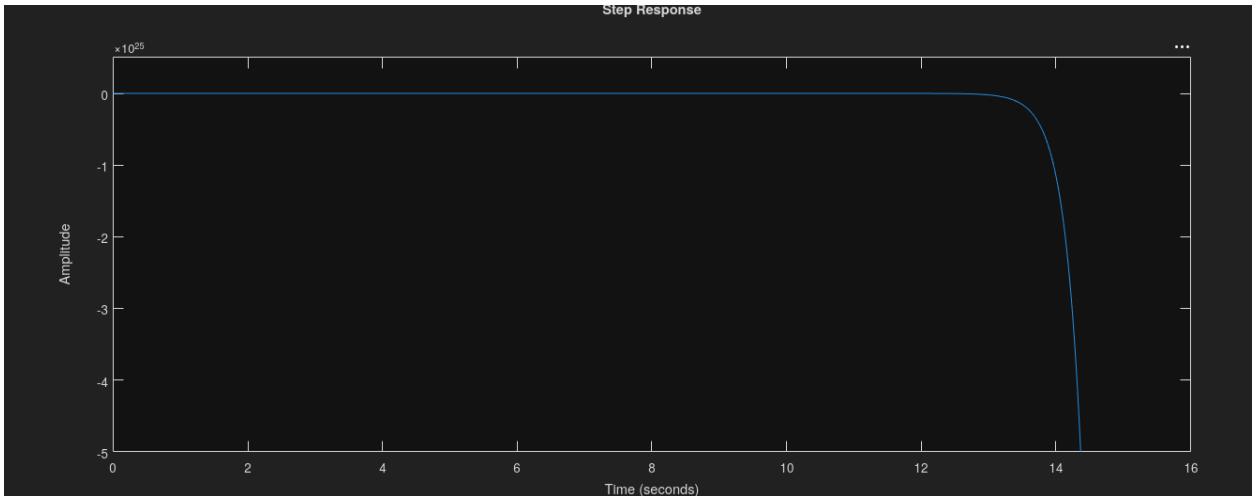
  s^2 - 18 s - 10
  ---------------
   s^2 - 5 s + 4

Continuous-time transfer function.
Model Properties
>> stepinfo(sys)
Warning: Simulation did not reach st
> In DynamicSystem/stepinfo>localSTE
In DynamicSystem/stepinfo (line 59)

ans =

  struct with fields:

        RiseTime: NaN
    TransientTime: NaN
     SettlingTime: NaN
      SettlingMin: NaN
      SettlingMax: NaN
        Overshoot: NaN
       Undershoot: NaN
             Peak: Inf
         PeakTime: Inf
```



e)

```
DEN =

     2   -23    -6

>> sys = tf(NUM,DEN)

sys =

  s^2 - 18 s - 10
  ---------------
  2 s^2 - 23 s - 6

Continuous-time transfer function.
Model Properties
>> stepinfo(sys)
Warning: Simulation did not reach steady sta
> In DynamicSystem/stepinfo>localSTEPINFO (
In DynamicSystem/stepinfo (line 59)

ans =

  struct with fields:

        RiseTime: NaN
    TransientTime: NaN
     SettlingTime: NaN
      SettlingMin: NaN
      SettlingMax: NaN
        Overshoot: NaN
       Undershoot: NaN
             Peak: Inf
         PeakTime: Inf

>> step(sys)
>>
```

